

Driver Drowsiness Detection using Transformer Architectures



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Introduction

Driver fatigue is a leading cause of accidents.

Existing CNN-based systems often fail in real-world conditions:

- Shadows or poor lighting
- Head movement or partial occlusion

Recent research shows Transformer-based models perform better in capturing long-range spatial dependencies

Goal: Build a deep learning model to classify eye states (open / closed) for real-time drowsiness detection

Existing Work

- Earlier studies mainly used **CNN-based models** for eye-state detection and drowsiness monitoring.
- CNN methods work well in controlled settings but **struggle with shadows, lighting changes, and head movements**.
- Research shows CNN's capture **only local features**, limiting their ability to detect subtle global facial cues.
- Transformer-based approaches have shown **higher accuracy and robustness** compared to traditional CNNs.
- Still, there is **limited research on real-time transformer-based drowsiness detection**, leaving room for improvement.

Dataset

MRL Eye Dataset (Kaggle)

- 84,898 images
- Open: 42,952 | Closed: 41,946
- Varying lighting, pose, glasses

CEW Dataset

- Real-world open/closed eye images
- High variation in lighting and angles

Purpose

- MRL → Training
- CEW → Real-world testing & generalization



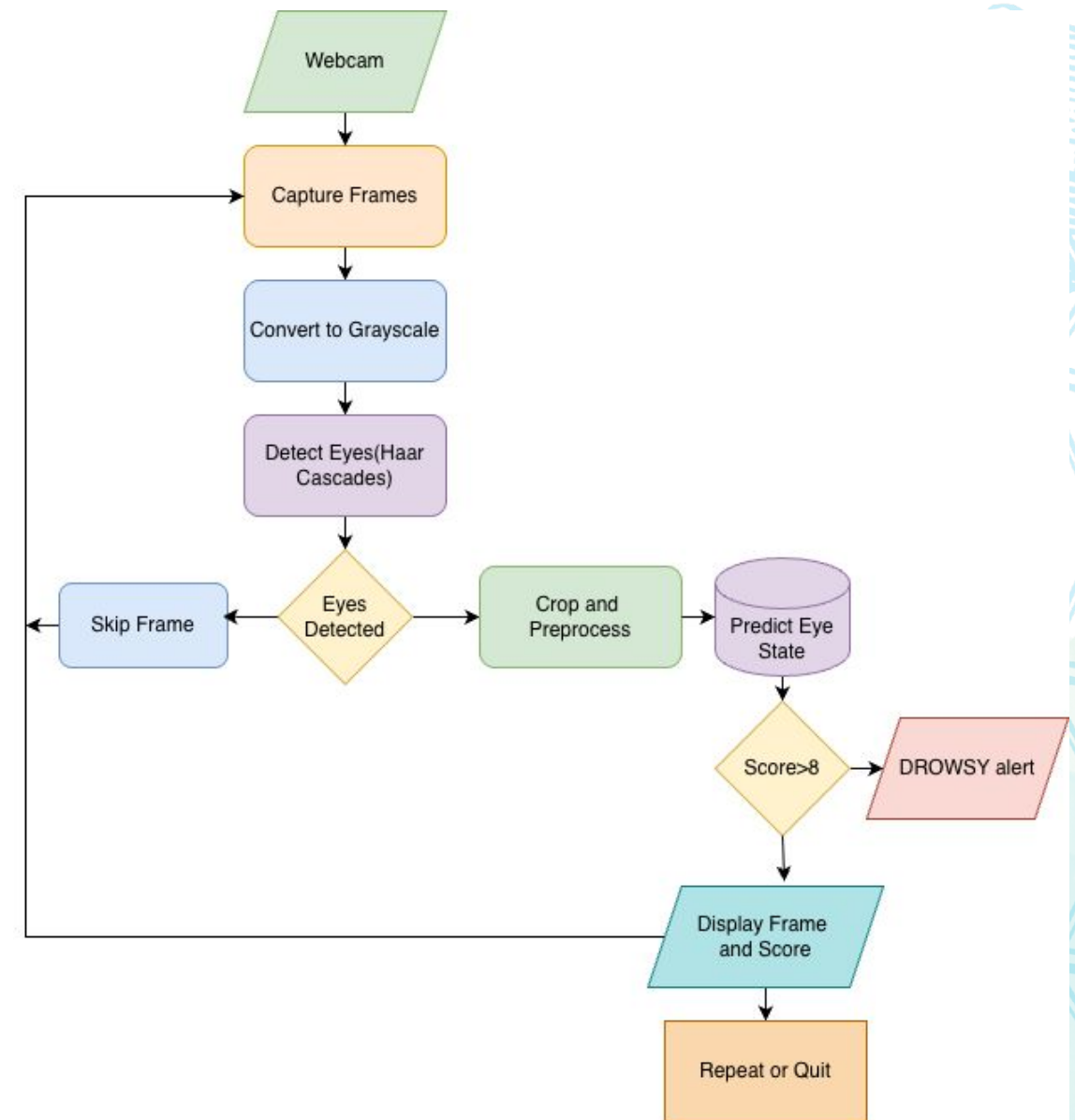
Methodology

Training: Use **MRL Eye Dataset** to train three models: ViT, Swin Transformer, and ResNet-50.

Cross-Dataset Testing: Evaluate all trained models on the **CEW dataset** to check generalization on unseen images.

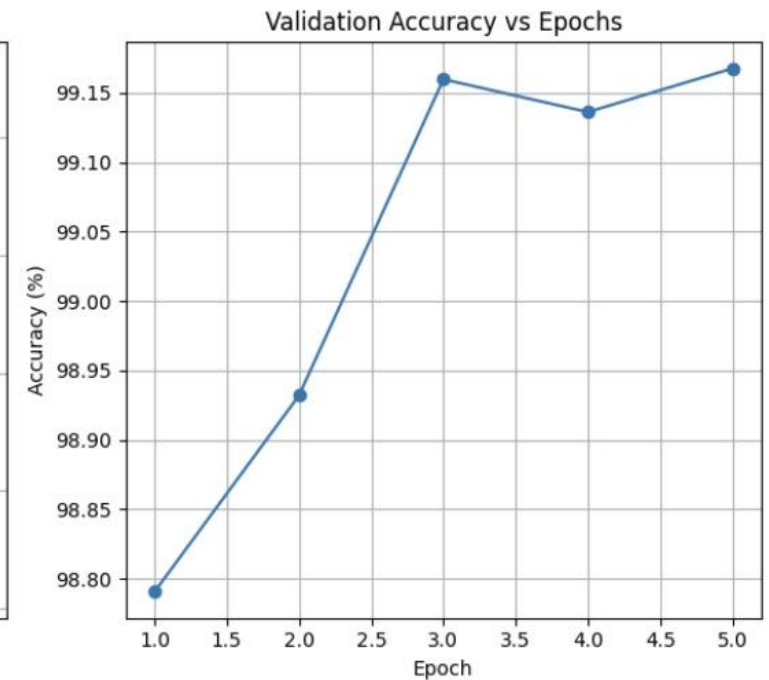
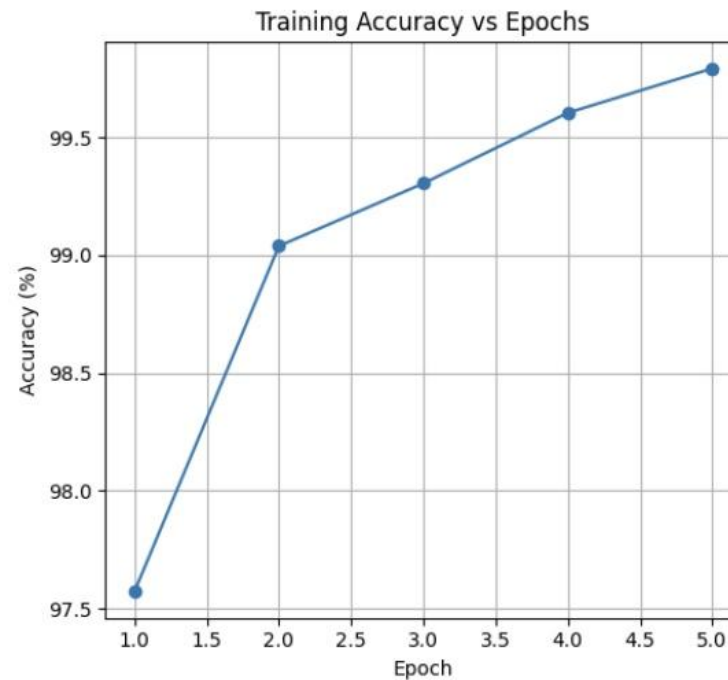
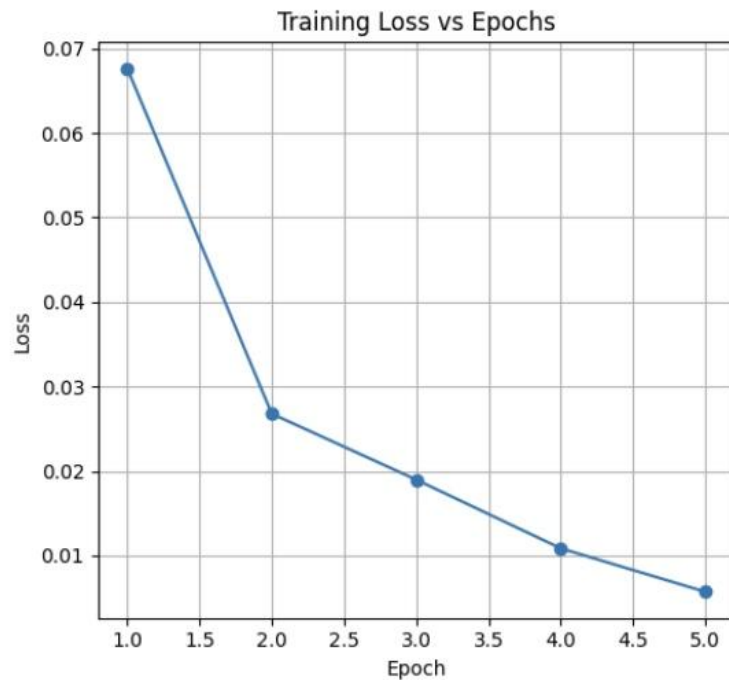
Real-Time Testing: Deploy the trained Transformer models to process live video of a team member, detecting eye state (open/closed) and identifying drowsiness when eyes remain closed for a fraction of a second.

Goal: Compare models' performance in both controlled datasets and real-world scenarios to find the most reliable solution.



Training Data Accuracy

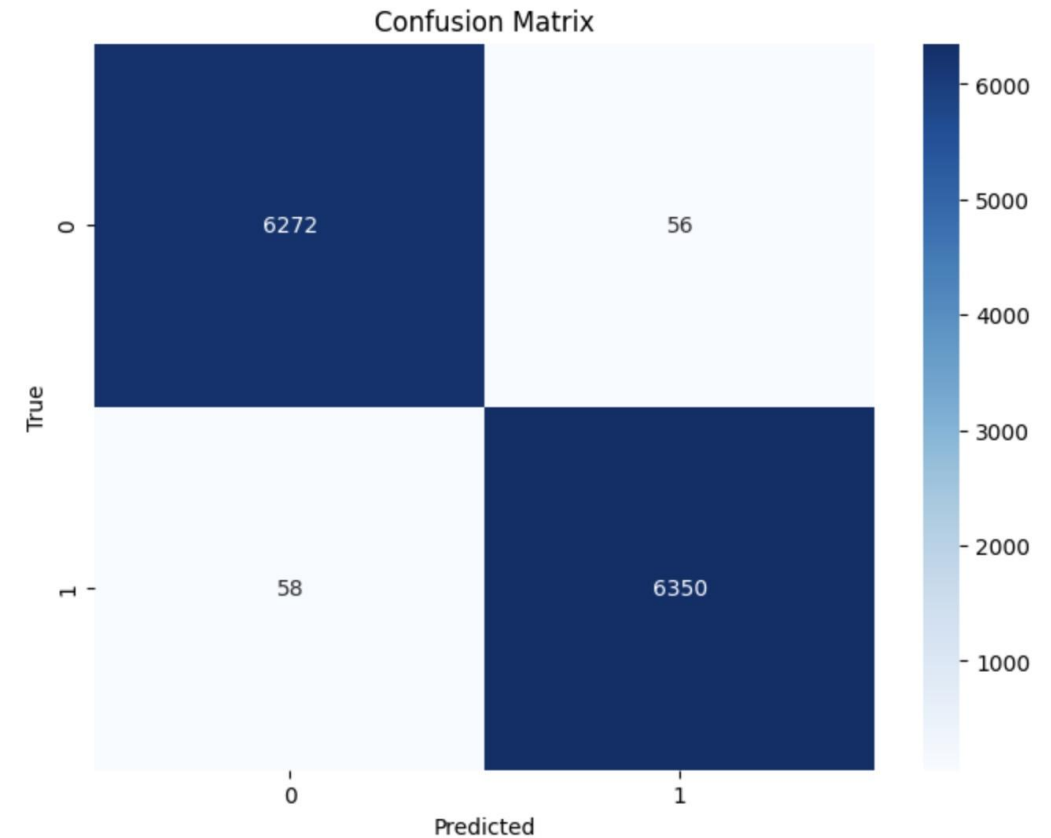
- Split the data into 70:15:15 ratio for training , validation and testing.



Test Data Accuracy

Classification Report:

	precision	recall	f1-score	support
0	0.99	0.99	0.99	6328
1	0.99	0.99	0.99	6408
accuracy			0.99	12736
macro avg	0.99	0.99	0.99	12736
weighted avg	0.99	0.99	0.99	12736



Test Accuracy on CEW Dataset

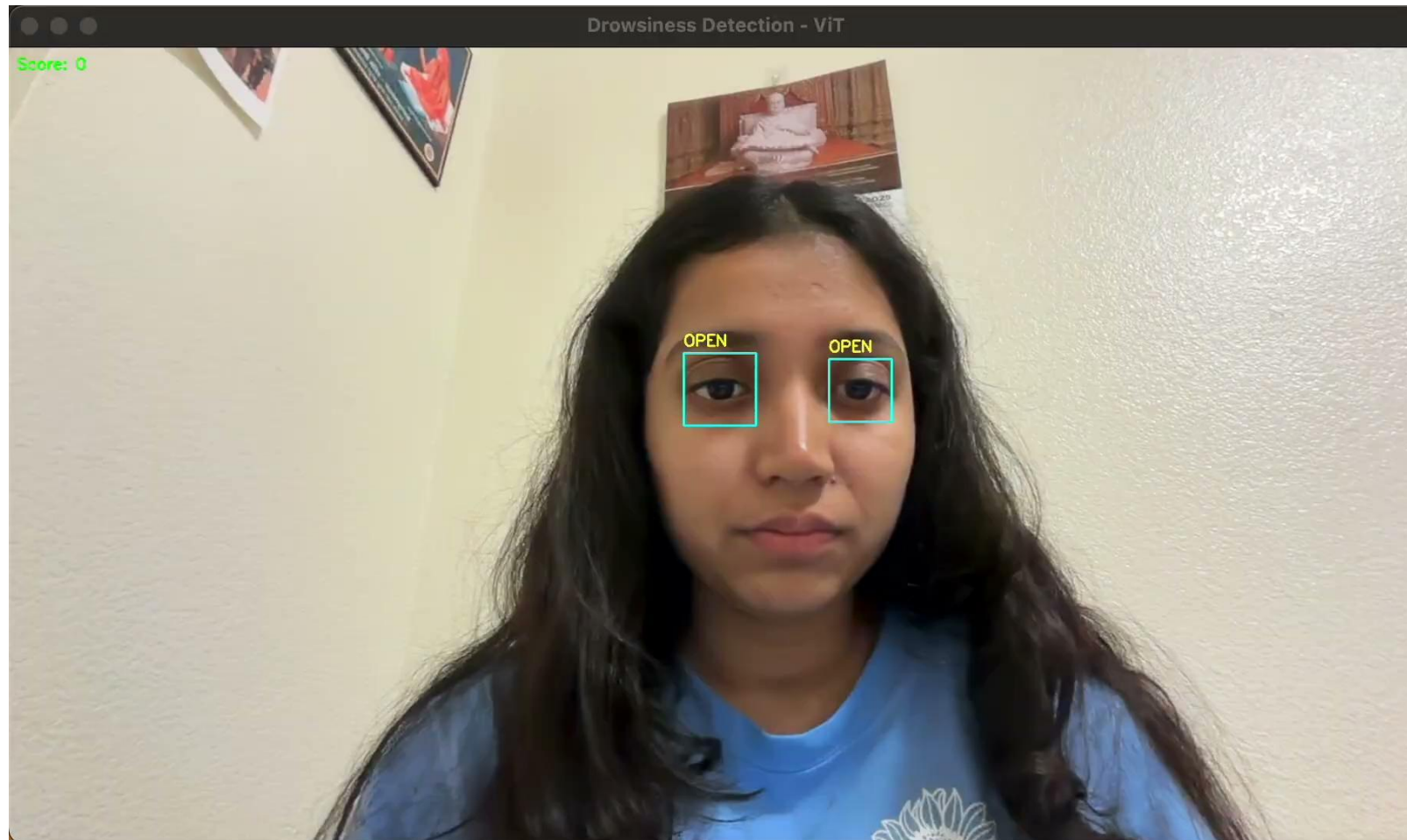
Model	Total Accuracy	Closed Face Precision	Open Face Precision
Swin-Transformer	92.53%	0.88	0.94
Resnet-50	74.16%	74.16%	0.76
ViT	93.49%	0.95	0.93

ViT emerged as the superior model, achieving the highest accuracy (93.49%) and minimizing false alarms with exceptional precision (0.95) for detecting closed eyes.

Conclusion

- The project aimed to develop a robust eye-state classifier for real-time drowsiness detection.
- Cross-dataset testing on CEW showed that Transformers—especially Swin—generalize better to real-world variations.
- Overall, Transformer architectures proved more reliable for real-time driver monitoring applications.
- Future work includes improving inference speed and deploying the system on embedded devices.

Demo





Thank you